KERALA SSLC | MATHEMATICS

SECOND DEGREE EQUATIONS

PYQ • 2021–2026

1. 2026 **3 MARKS**

One side of a rectangle is 30 metres longer than the other and its area is 351 square metres.

(i) Write the above information as a second-degree equation. **(1)**

(ii) Calculate the length of the shorter side. **(2)**

OR

2. 2026 **3 MARKS**

Find two numbers with sum 4 and product 2.

3. 2026 **4 MARKS**

The sum of the first n terms of an arithmetic sequence is $n^2 + 6n$.

(i) What is the sum of the first 7 terms? **(2)**

(ii) How many terms of this sequence, starting from the first, must be added to get 315? **(2)**

Application-based question: Arithmetic Sequences leading to a second-degree equation.

4. 2025 **3 MARKS**

Six times a natural number subtracted from the square of that number gives 187.

(a) Form a second-degree equation by taking the number as x.

(b) Find the number.

5. 2024 **3 MARKS**

Ajay is 10 years older than Renuka. The product of their ages is 144.

(a) Taking the age of Renuka as x, what is the age of Ajay in terms of x?

(b) Find their ages.

6. 2024 **3 MARKS**

The square of a number is equal to 12 added to the number. Find the number.

7. 2023 **3 MARKS**

Find the lengths of the sides of the rectangle whose perimeter is 80 centimetres and area 351 square centimetres.

8. 2023 **4 MARKS**

In a right triangle, one of the perpendicular sides is 2 centimetres more than that of the other. Area of the triangle is 24 square centimetres. Find the lengths of the perpendicular sides of the right triangle.

9. 2022 **4 MARKS**

A strip of width 4 centimetres is attached to one side of a square to form a rectangle. The area of the new rectangle is 77 square centimetres.

(a) If we take the width of the new rectangle as x, what will be its length?

(b) Find the measure of the side of the square by constructing an equation.

10. 2022 **6 MARKS**

The product of a number and 5 more than that number gives 104.

(a) If we take the first number as 'r', what will be the second number?

(b) Form a second degree equation using the given details. **(2)**

(c) Find the number.

11. 2021 **3 MARKS**

(a) Write the sequence of even numbers. **(1)**

(b) One added to the product of two consecutive even numbers gives 289. Form a second degree equation to solve this problem. **(2)**

12. 2021 **4 MARKS**

(a) The perimeter of a rectangle is 40 centimetres. Length of its smaller side is 7 centimetres. What is the length of its larger side? **(1)**

(b) Find the sides of a rectangle with perimeter 40 centimetres and area 96 square centimetres.